

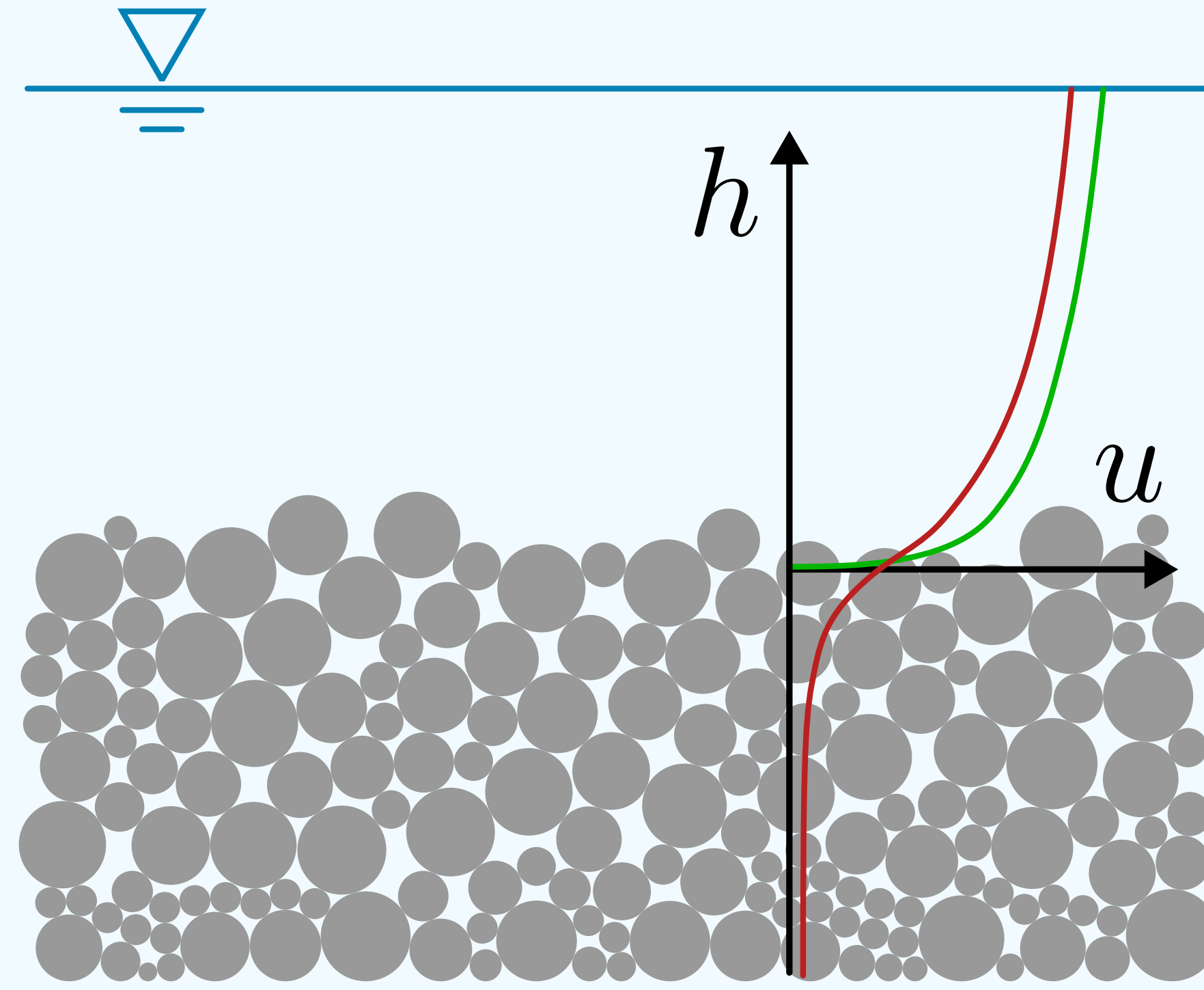
Hyporheic flow and velocity distribution

A semi-analytical approach

Context and Objective

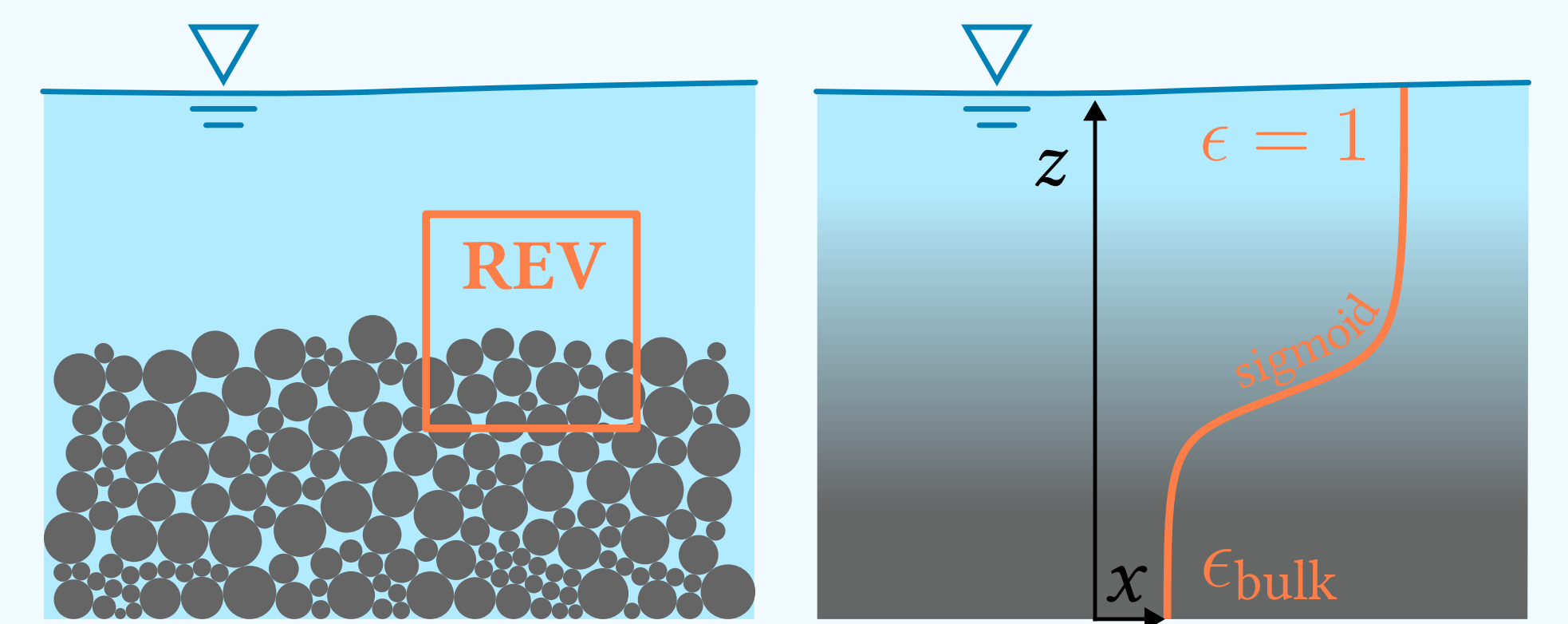
A river is usually modeled as a closed system, with the bed constituting the bottom boundary of the domain. However, when the bed is porous, the hyporheic (subsurface) flow exchanges momentum with the main flow and governs the velocity profile.

Building on established empirical relationships that illustrate the velocity profile of the main flow, we aim to incorporate hyporheic flow explicitly. By closing the Reynolds and *dispersive* stresses, along with the drag force density for flows over porous beds (Rousseau & Ancey, 2022), one can derive the one-dimensional momentum equation governing the vertical velocity profile. A numerical solver is then implemented to compare the solution with the existing empirical friction laws.



Introducing porosity

Because the bed has an irregular shape and disrupts the uniformity of the flow, we need to observe the flow at a scale larger than the particle size. We perform a spatial average over a representative elementary volume (REV) to enable the inclusion of porosity effects in the Navier-Stokes equations (Nikora, 2001). An additional stress, known as dispersive stress, emerges. This stress was calibrated with a small-scale experiment (image on the right ; Gauthier & Ancey, 2022) as a mixing length with a single parameter $\lambda = 0.015$.



$$\mathbf{p}'_x = \epsilon \rho g i + \frac{d}{dz} (\tau_v + \tau_t + \tau_d) + \epsilon f - \mu \frac{du}{dz} \frac{d\epsilon}{dz} = 0$$

$$\tau_d = -\rho \epsilon \langle \tilde{u}_x \tilde{u}_z \rangle$$

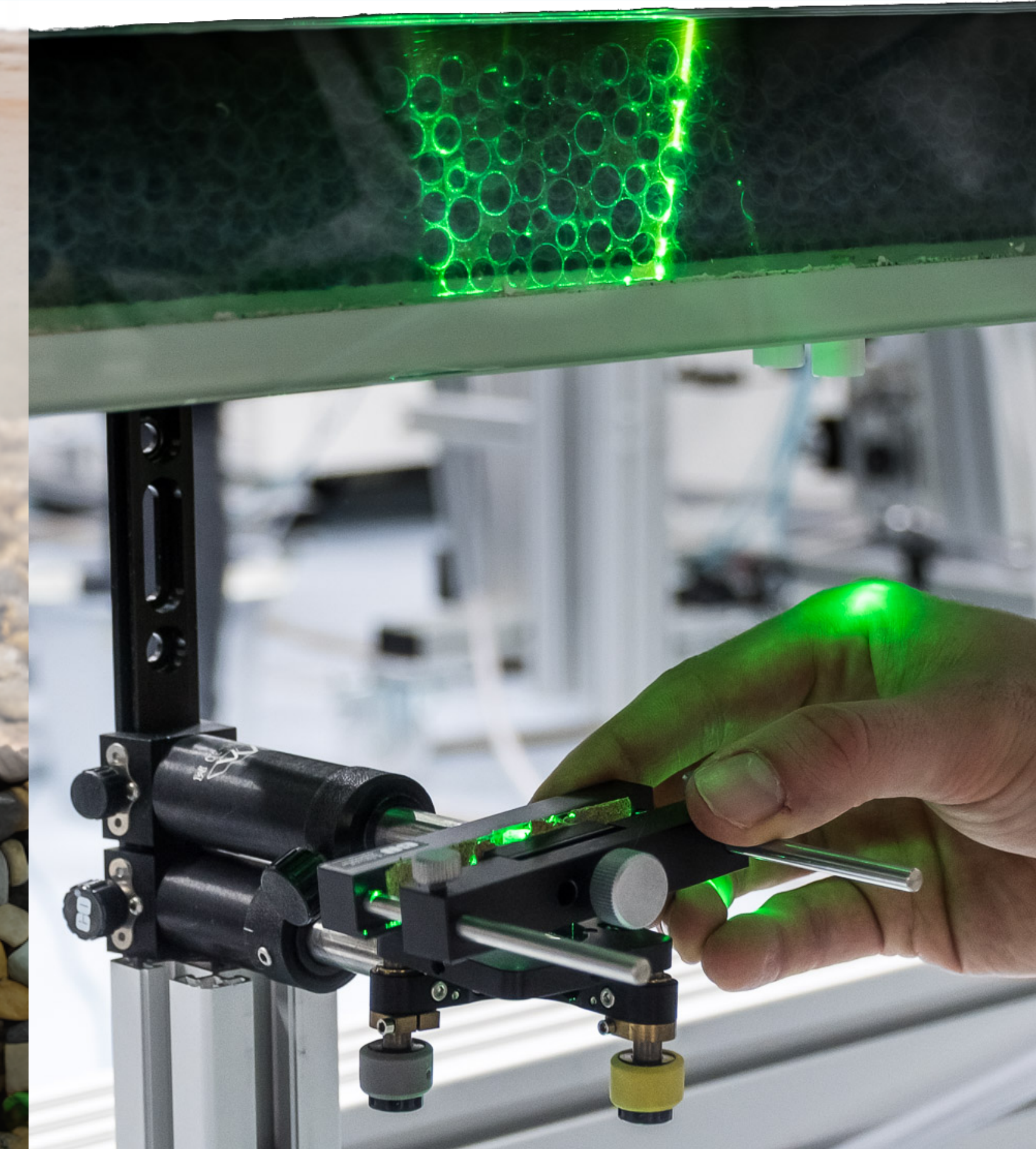
$$0 = c_0 + c_1 u' + c_2 u'^2 + c_3 u'^3 + c_4 u' u'' + c_5 u'', \quad c_i = c_i(z, u)$$

$$\tau_t = \rho \epsilon \kappa^2 Z_{LS}^2 \left(1 - e^{-\sqrt{\frac{Z_{LS} u}{v A_*^2}}} \right)^2 \left(\frac{du}{dz} \right)^2$$

$$\tau_d = \rho \epsilon \lambda d_p \frac{1 - \epsilon}{1 - \epsilon_b} u \frac{du}{dz}$$

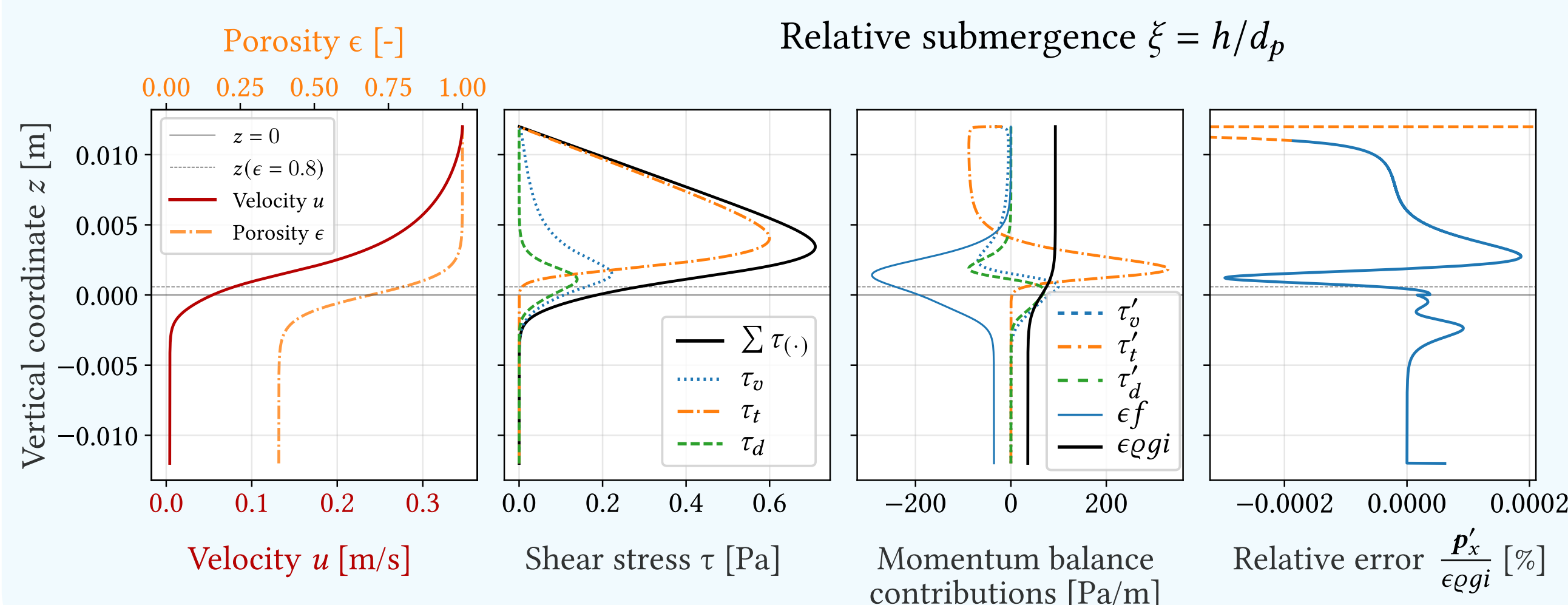
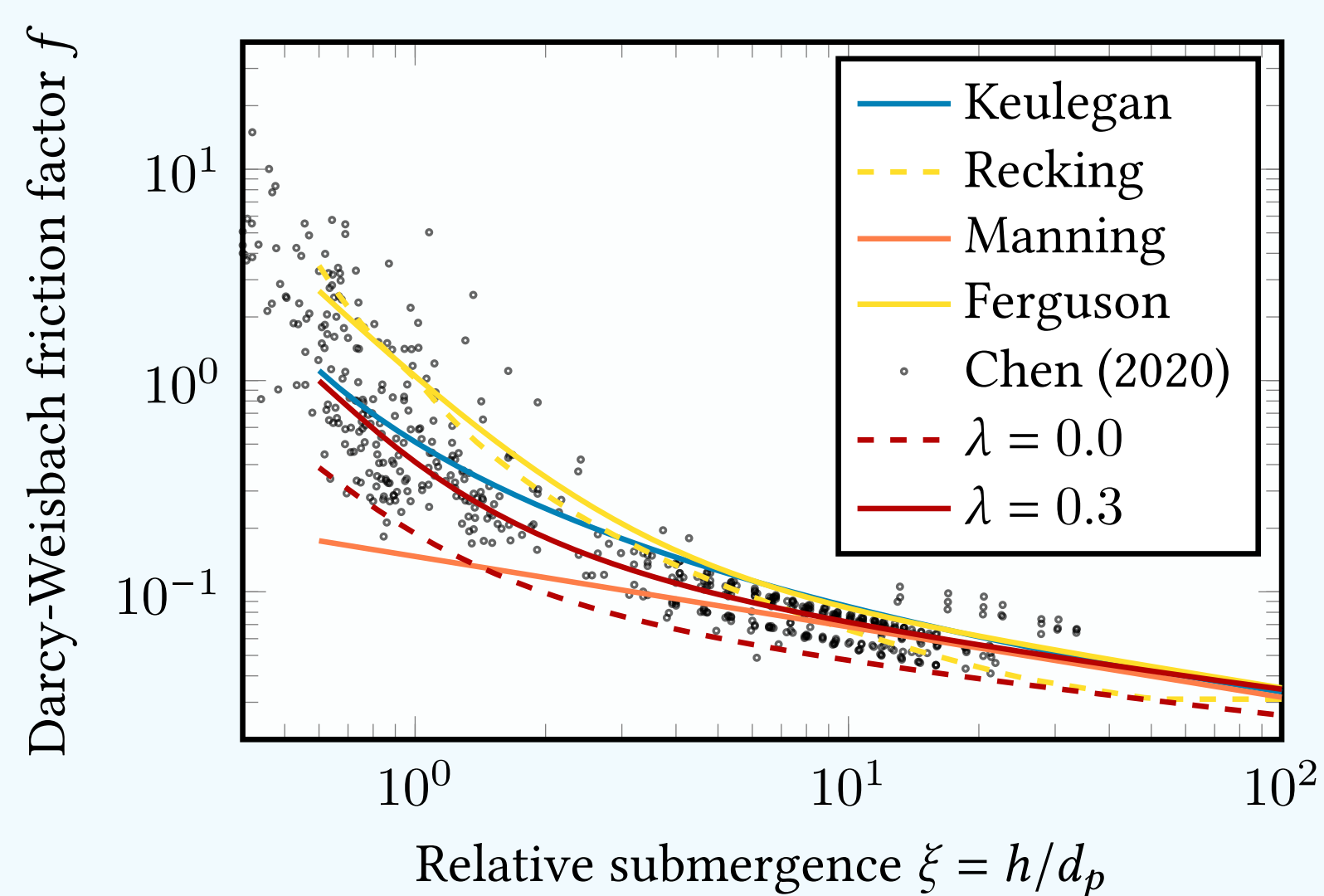
$$f = -\rho v A_E \left(\frac{1 - \epsilon}{\epsilon d_p} \right)^2 u - \rho B_E \frac{1 - \epsilon}{d_p} u^2$$

Credits : Bob de Graffenried



Numerical solution

By converting our second-order ordinary differential equation into a linearized partial differential equation (by adding a residual), we can use the finite element method to solve the equation numerically. At the bottom boundary, a simple Darcy velocity is set, while the top boundary is left free of any traction.



Discussion & Conclusion

The numerical solution accurately represents the overall behavior of very shallow flows, with the parameter λ adjusted to 0.3 to align with the established empirical laws in the limit of deep flow, where $h / d_p \rightarrow \infty$. The high value of the parameter λ (relatively to its experimental value of 0.015) highlights the important role that the dispersive stress plays in the velocity profile. Setting $\lambda = 0$ removes the agreement entirely. This gap is itself a result: it indicates the importance of the dispersive stress in the momentum balance.

Next step

The next step is to solve the equation analytically. We will examine the same equation by deriving matched asymptotic expansions, with distinct inner and outer regions within and above the bed. This approach aims to obtain an approximate velocity profile and reveal the parameter groupings that the numerical solution cannot.

References

- Nikora, Vladimir et al. "Spatially Averaged Open-Channel Flow over Rough Bed". In: Journal of Hydraulic Engineering 127.2 (2001), pp. 123–133. doi: 10.1061/(ASCE)0733-9429(2001)127:2(123).
- Rousseau, G. and C. Ancey. "An experimental investigation of turbulent free-surface flows over a steep permeable bed". In: Journal of Fluid Mechanics 941 (2022), A51. doi: 10.1017/jfm.2022.310.